

# Spatial scRNA-seq

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# Overview

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# 1. Abstract

## 2. Introduction

### 3. Materials

## 4. Methods

# Methods I

## Basic method

- 1 Scanpy (Wolf, Angerer, & Theis, 2018)

## Projection method

- 1 UMAP (McInnes, Healy, & Melville, 2018; Becht et al., 2019)
- 2 tSNE (Maaten & Hinton, 2008)
- 3 PHATE (Moon et al., 2019)
- 4 trimap (Amid & Warmuth, 2019)

## Clustering

- 1 Leiden (Traag, Waltman, & van Eck, 2019)
- 2 Louvain (Blondel, Guillaume, Lambiotte, & Lefebvre, 2008)

## 5. Results

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### 5.1. Quality check

# Total counts per cell I

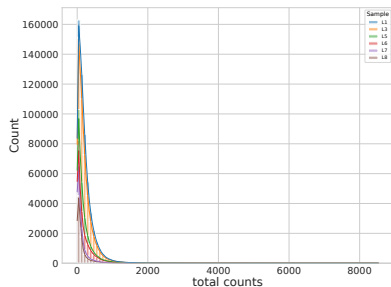


Figure: Total counts

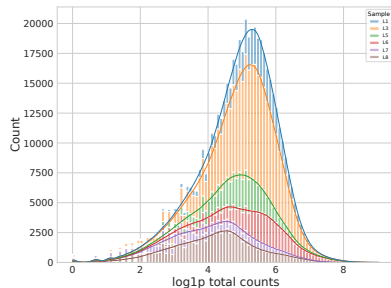


Figure:  $\log_{1p}(\text{Total count})$

# Total counts per cell II

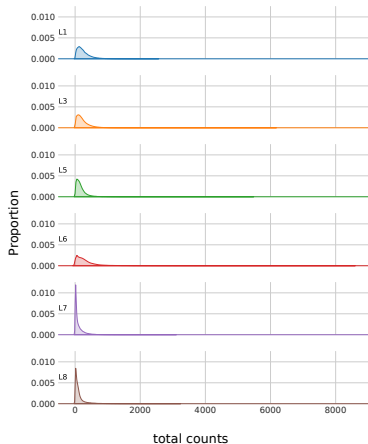


Figure: Genes by counts

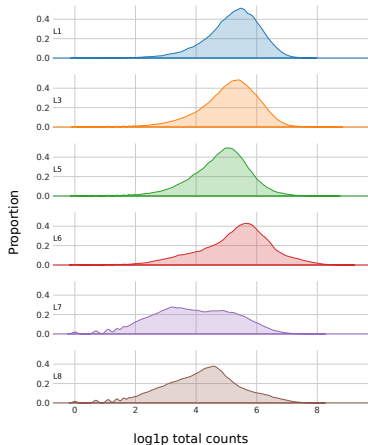


Figure: log1p(Genes by count)

# Total counts per cell III

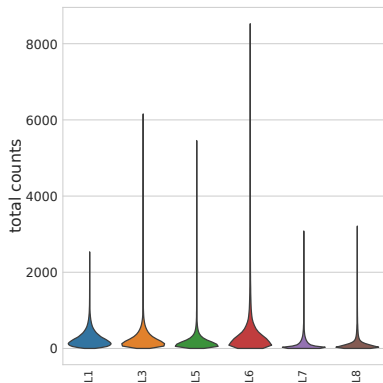


Figure: Genes by counts

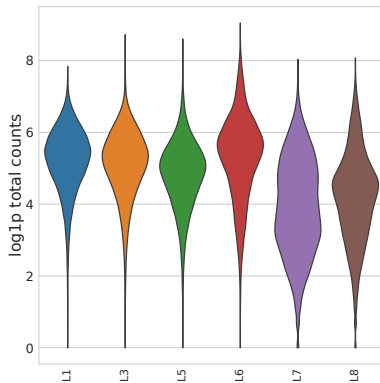


Figure: log1p(Genes by count)

# Genes by counts I

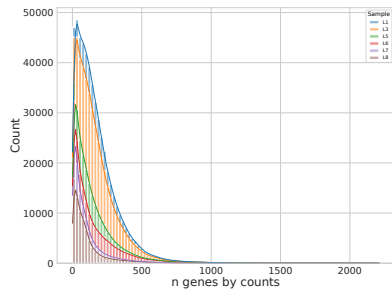


Figure: Genes by counts

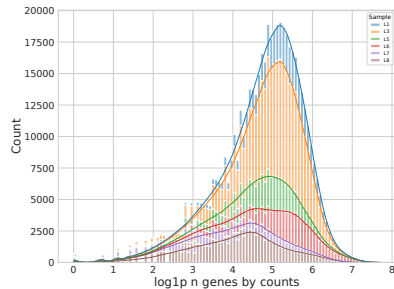


Figure:  $\log_{1p}(\text{Genes by count})$

# Genes by counts II

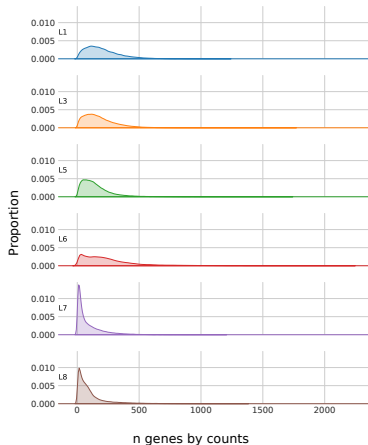


Figure: Genes by counts

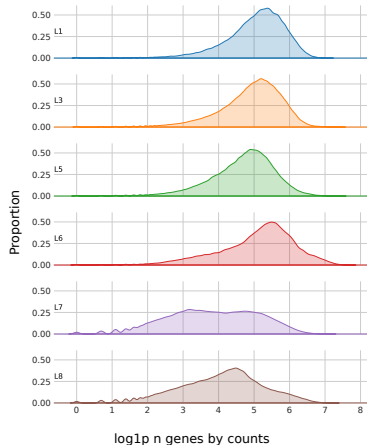


Figure:  $\log_{1p}(\text{Genes by count})$

# Genes by counts III

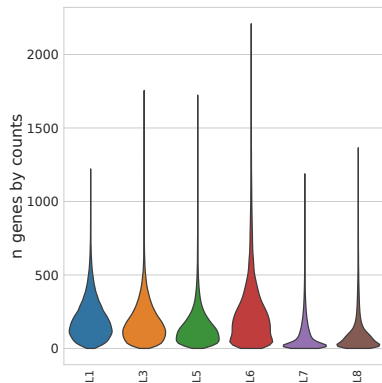


Figure: Genes by counts

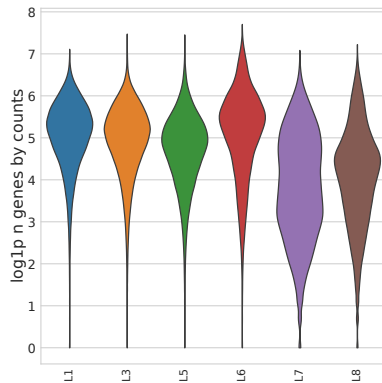


Figure: log1p(Genes by count)

## 5. Results

### 5.2. Highly variable genes

# Highly variable genes I

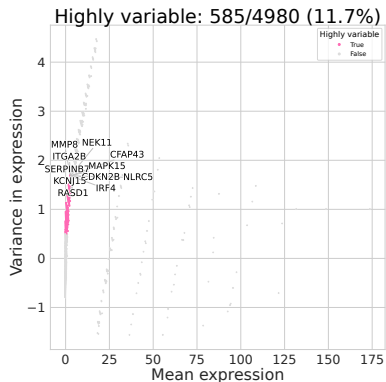


Figure: Volcano plot

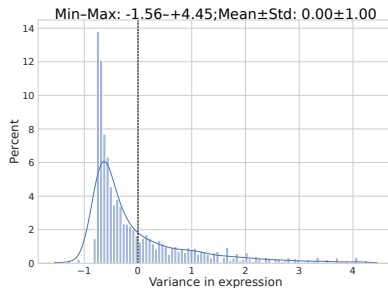


Figure: Histogram

# Highly variable genes II

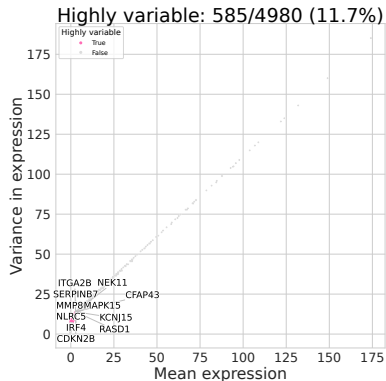


Figure: Volcano plot

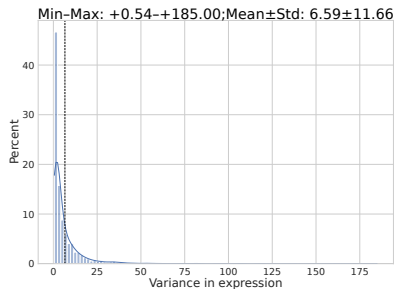


Figure: Histogram

## 5. Results

### 5.3. PCA

## 5. Results

### 5.4. Clustering

# Clustering with UMAP I

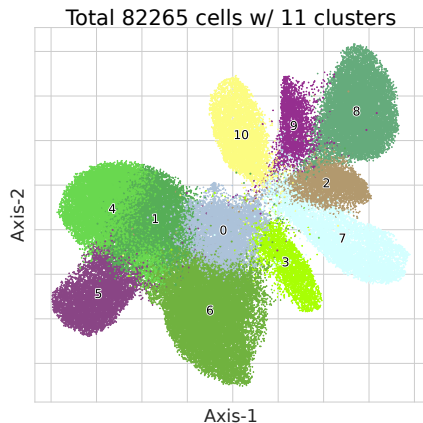


Figure: Clustering with UMAP

## 5. Results

### 5.5. Marker genes

## 6. References

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